

Reading the GNoME Oxide Screen

A column-by-column reference for the two oxide CSVs

`15_filter_oxides.py` reads the full GNoME screen output — `gnome_screen_all.csv`, 33,323 candidates scored by `13_screen_gnome.py` — and pulls out every row whose formula contains oxygen, writing two files (both in `results/cgcnn/`):

File	Rows
<code>gnome_oxide_candidates.csv</code>	11,612 — every scored oxide
<code>gnome_oxide_low_kappa_candidates.csv</code>	1,545 — oxides with $\kappa_L \leq 1 \text{ W m}^{-1} \text{ K}^{-1}$

Both share the same columns (the low- κ file drops `is_low_kappa_candidate`, since every row there is one by definition). This document explains what each column means.

Identity

Column	Meaning
<code>material_id</code>	GNoME's own identifier for the material.
<code>formula</code>	Reduced chemical formula, e.g. <code>SbW04</code> .
<code>stoichiometry_pattern</code>	A rough structural family, read off the <i>reduced</i> cation:oxygen ratio — see the box below.
<code>is_low_kappa_candidate</code>	(<code>gnome_oxide_candidates.csv</code> only) True if this row also appears in the low- κ file.

What `stoichiometry_pattern` means

A pattern-match on the *ratio* of cations to oxygen after reducing to lowest integers — not a verified crystal structure, which would need atomic positions this column does not look at. **ABO3-type**: two cations in 1:1 ratio, three oxygens (perovskite-like). **AB2O4-type**: cations in 1:2, four oxygens (spinel-like). **A2B2O7-type**: cations in 1:1 (reduces from 2:2), seven oxygens (pyrochlore-like). Anything else with two cations is labelled a generic ternary oxide; three or more cations become *complex oxide* (*n elements*). In this particular screen no exact ABO3 or simple binary (single-cation) oxide survived the paper's own bandgap/stability/ non-radioactive filters — GNoME's discovered materials skew toward more complex, previously-uncatalogued compositions, and simple perovskites are mostly already well known.

Structural quantities

Read directly from the primitive cell of the parsed structure — not predicted, not uncertain.

Column	Meaning
<code>Number of Atoms</code>	Atom count in the <i>primitive</i> cell (not whatever supercell GNoME's CIF happened to use).
<code>Volume (Å³)</code>	Primitive-cell volume, Å ³ .
<code>Density (g cm⁻³)</code>	Mass ÷ volume.
<code>Atomic mass (amu)</code>	Total composition weight of the primitive cell.

Model predictions

The actual machine-learning output — the mean of the 3-model CGCNN ensemble, in linear GPa (converted back from the $\log_{10}$ space the network is trained in).

Column	Meaning
K_VRH_pred	Predicted bulk modulus K (GPa).
G_VRH_pred	Predicted shear modulus G (GPa).
K_VRH_spread_log10	How much the 3 ensemble members <i>disagree</i> on K , in $\log_{10}(\text{GPa})$ — a per-crystal uncertainty estimate, not a physical property.
G_VRH_spread_log10	Same disagreement measure for G .

Derived physics (the Slack model)

Everything below is computed from K and G above via the paper’s unchanged Slack-model formulas (`scripts/cgcnn/07_predict_kappa.py`, `slack_physics()`).

Column	Meaning
Poisson ratio	From the ratio of longitudinal to transverse sound velocity, $v_{\text{long}} = \sqrt{(K + \frac{4}{3}G)/\rho}$, $v_{\text{trans}} = \sqrt{G/\rho}$.
Gruneisen parameter	How anharmonic the lattice vibrations are, derived from the Poisson ratio. Larger $\Rightarrow$ phonons scatter off each other more strongly $\Rightarrow$ generally lower κ_L .
Kappa_cal (W m ⁻¹ K ⁻¹)	The predicted lattice thermal conductivity — the number this whole screen thresholds on.

$$\kappa_L = \frac{G \cdot v_{\text{sound}} \cdot V^{1/3}}{N_{\text{atoms}} \cdot 300} \cdot e^{-\gamma} \quad (\text{Eq. 2 of the PINK paper; } \delta = 1)$$

Uncertainty interval (Monte Carlo)

Column	Meaning
Kappa_cal_p05	5th percentile of κ_L across 2,000 Monte Carlo draws.
Kappa_cal_p50	Median of the same 2,000 draws — sits close to Kappa_cal above.
Kappa_cal_p95	95th percentile.

How the interval is built, and how to read it

K and G are each redrawn 2,000 times from a normal distribution centred on the ensemble prediction, with width equal to $\text{K_VRH_spread_log10} / \text{G_VRH_spread_log10}$ — i.e. how much the 3 ensemble members disagreed. Structural quantities stay fixed; only the *predicted* moduli are uncertain. Every draw goes through the identical Slack-model formula used for the point estimate, so p05/p50/p95 are just percentiles of the resulting 2,000 κ_L values — consistent with Kappa_cal by construction, not a separate approximation. **A tight p05–p95 range means the ensemble agreed and the prediction is trustworthy; a wide range means the 3 models disagreed and the candidate is worth a real DFT check before believing it.** This is something a single point-estimate model (like the paper’s own released checkpoint) cannot report at all.